BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Network Threats

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. How can more active users affect network performance?

2. How does a brute-force attack work?

A) Tries many possible passwords
B) Sends a trusted-looking message
C) Floods a web server
D) Rearranges file fragments

3. "A MAC address is the text domain name entered into a browser." Correct this. Explain why.

4. Data must reach another network in the right order. "TCP chooses fonts; IP draws images." Correct this. Explain each protocol's job.

Today's objectives

Explain the core ideas of network threats using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Malware
Malicious software such as ransomware, spyware or viruses.

Social engineering
Manipulating people into revealing information or bypassing safeguards.

Denial of service
Overwhelms a service to degrade or prevent access.

Unauthorised access
Guessing credentials or exploiting flaws may expose or change protected information.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Malware	Malicious software such as ransomware, spyware or viruses.
Social engineering	Manipulating people into revealing information or bypassing safeguards.
Denial of service	Overwhelms a service to degrade or prevent access.
Unauthorised access	Guessing credentials or exploiting flaws may expose or change protected information.

I Do - One worked example

A staff member receives a message asking for their password through an unfamiliar link. Identify the risk and response.

1. Classify the attempt as phishing/social engineering.
2. Explain that the link could harvest credentials.
3. Do not enter details; verify independently via official channels.
4. Report to school IT and follow account-security procedures.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. An online service receives huge volumes of requests and slows down. Diagnose.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the line that increments an observed failed-login count.

```
attempts = 2
...
print(attempts)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Give the meaning of phishing. [2 marks]

Q3. Explain how ransomware can harm a school. [3 marks]

Q4. Name one precaution against social engineering. [2 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Phishing is a form of social engineering. ☐ T ☐ F
2. Malicious software such as ransomware, spyware or viruses. ☐ T ☐ F
3. Manipulating people into revealing information or bypassing safeguards. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Users may compete for finite bandwidth, reducing the data rate available to each. ☐

Q2 A - Tries many possible passwords ☐

Q3 A MAC address identifies a network interface on a LAN. It is usually six pairs of hexadecimal digits. ☐

Q4 TCP supports reliable delivery in order. IP handles addressing and routing between networks. Each protocol has its own job; encryption does not prove a site is honest. ☐

Code First - Model

```
attempts = 2
attempts = attempts + 1
print(attempts)
```

Output: 3

We Do - Check the model

1. Distinguish attack evidence from a normal traffic spike. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Deceptive communication designed to trick a user into disclosing data or taking unsafe actions. ☐

Q3 Malware may encrypt/deny access to files, disrupt services and cause recovery costs. ☐

Q4 Staff training, verifying requests independently or multifactor authentication. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: